SAKSHAM TIMALSINA

University of Wyoming
Department of Civil and Architectural Engineering
and Construction Management

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RESEARCH INTERESTS

Energy infrastructure safety and resilience, onstruction disputes and resolutions, construction safety, laser scanning, hydrogen energy, hydrogen transportation infrastructure, nuclear energy, and machine learning.

EDUCATION

University of Wyoming — Laramie, WY
Ph. D. Student, Civil Engineering — Expected: May 2027
Advisor: Dr. Chengyi Zhang

Tribhuvan University — Kathmandu, Nepal
B.S. in Civil Engineering — Sep 2022
Thesis: "*Flood Inundation Mapping and Dam Breach Analysis of Mugu-Karnali Basin using HEC-RAS*"

RESEARCH EXPERIENCE

University of Wyoming — Laramie, WY
Research Assistant (with Prof. Chengyi Zhang) — Jan 2024- present

- Application of both terrestrial and aerial laser scanners in the 3D model development, inspection, and maintenance of infrastructures.
- Hydrogen transportation infrastructure in Wyoming. Studied the relevance of different hydrogen infrastructures, including blending into natural gas pipelines, trucks, etc.
- Disputes in hydrogen pipelines and resolution through collaborative efforts and their prediction using machine learning models.

Tribhuvan University — Kathmandu, Nepal
Undergraduate Researcher — May 2021 – May 2022

- Water resources and disaster risk analysis through inundation mapping and dam breach analysis of Mugu-Karnali River segment using GIS and HEC-RAS.

TEACHING EXPERIENCE

Construction Documents, University of Wyoming — Jan-May 2026

- Grading, class assistance, and office hours

Introduction to Construction Management, University of Wyoming — Aug-Dec 2025

- Grading, office hours

Building Information Modelling, University of Wyoming — Jan-May 2025

- Grading, lab assistance, and office hours

Engineering Economics, University of Wyoming — Jan-May 2025

- Grading, office hours, and took 2 classes independently

Introduction to Civil, Architectural Engineering, and Construction Management, University of Wyoming — Aug-Dec 2024

- Grading, office hours

ACADEMIC GRANTS/HONORS

- College of Engineering and Physical Sciences Travel Support Grant, 2025
- School of Graduate Education (SGE) Graduate Student Travel Award, 2025 and 2026
- Ranked in the top 250 (among 13,000 students) in the entrance exam by the Institute of Engineering, Nepal

PROFESSIONAL AFFILIATIONS AND REVIEW ACTIVITIES

- ASCE Student Member, American Society of Civil Engineers (ASCE)
- Reviewer for the Journal of Legal Affairs and Dispute Resolution in Engineering and Construction and Journal of Pipeline Systems Engineering and Practice, ASCE.
- Reviewer for the International Journal of Hydrogen Energy and Materials Today Sustainability, Elsevier.

INDUSTRY EXPERIENCE

Civil Engineering Intern — Laramie, WY
DCR Consulting & Engineering — May 2025- Aug 2025

- Preparation of updated drawings as per the field condition, grade design, and as-built drawings using AutoCAD, Civil3D, and Revit.
- Submittal registrar preparation and submittal review.
- Report generation and updates in P6.

Sub-engineer — Sindhupalchowk, Nepal
Sunkoshi Rural Municipality — Dec 2022- Nov 2023

- Design, estimate, and quality inspection of different civil engineering projects.

PUBLICATIONS

Journal papers (appeared/under review/in preparation)

1. Zhao, Z., Kumar, D., Zhang, C.*, Li, H., **Timalsina, S.** (2024) "Techno-economic analysis of green hydrogen integration into existing pipeline infrastructure: A case study of Wyoming." *International Journal of Hydrogen Energy.* Volume 93, Pages 574-584, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2024.10.441
2. **Timalsina, S.**, Zhang, C.*, Kumar, D., Holubnyak, E. (2025). "Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis." *International Journal of Hydrogen Energy.* Volume 111, Pages 635-647, ISSN 0360-3199, https://doi.org/10.1016/j.ijhydene.2025.02.336
3. Pal, U. K., Zhang, C.*, and **Timalsina, S.** (2025). "Comparative Analysis of Regulatory Compliance and Legal Issues in Small Modular Nuclear Reactor Deployment." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1270
4. **Timalsina, S.**, Zhang, C.*, Hada, S., Demirkesen, S. (2025). "Assessing Environmental, Safety, Regulatory, and Dispute Resolution Challenges in Hydrogen Pipeline Infrastructure." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1272.
5. **Timalsina, S.**, Zhang, C.*, Quapp, U., Demirkesen, S., Olorunnishola, A. (2025). "Machine Learning-Based Predictive Model for Dispute Occurrence and Resolution Strategies in Pipeline

Projects." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction.* American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1334

6. Giri, M., Zhang, C.*, **Timalsina, S.**, Hada, S., Holubnyak, E. (2025). "A Comparative study of Regulatory and Policy Framework in Hydrogen Pipeline Development." *Journal of Legal Affairs and Dispute Resolution in Engineering and Construction*. American Society of Civil Engineers. https://doi.org/10.1061/JLADAH.LADR-1356
7. **Timalsina, S.**, Zhang, C.*, Pal, U. K. (2026) "Quantifying reliability and uncertainty in hydrogen infrastructure through integrated incident analysis." Journal of Loss Prevention in the Process Industries. Volume 100, 105921, ISSN 0950-4230, https://doi.org/10.1016/j.jlp.2026.105921

Conference papers

1. Mahajan, D., Sadikoglu, E., Pal, U. K., **Timalsina, S.**, Zhang, C.*, and Demirkesen, S. (2025). A Multi-Output Regression Approach to Predicting Gender-Centric Workplace Fatal Accidents. *Journal of Engineering, Project, and Production Management*, 15(1), 0002. DOI: https://doi.org/10.32738/JEPPM-2025-0002

CONFERENCE AND PRESENTATIONS

1. Attended *Construction Safety Conference*, Lawrence, KS, 2024 representing "A Multi-Output Regression Approach to Predicting Gender-Centric Workplace Fatal Accidents".
2. Presented at *Wyoming's Energy Future Symposium*, Laramie, WY, 2024: Poster title: "*Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis*".
3. Presented at *Graduate Poster Symposium*, University of Wyoming, WY, USA, 2024: Poster title: "*Optimizing Hydrogen Transportation Infrastructure and Supply Chain under Uncertainty: An Integrated Mixed Integer Linear Programming and Real Options Analysis*".
4. Presented at *CI Summit 2025,* Costa Mesa, CA, 2025. Presentation title: *"Assessing Environmental, Safety, Regulatory, and Dispute Resolution Challenges in Hydrogen Pipeline Infrastructure".*

WORKSHOPS ATTENDED

1. *AI Unlocked: Empowering Higher Education through Research and Discovery* — Denver, CO, Apr 2-3, 2025
   National Artificial Intelligence Research Resource (NAIRR) Pilot Workshop
2. *Urban Building Energy Modeling Workshop* — Laramie, WY, Apr 29, 2025
   University of Wyoming

CERTIFICATIONS AND LICENSURES

1. OSHA 30-Hour Outreach Training
2. Procore Certification: Student
3. Federal Aviation Administration (FAA) Part 107 License for small Unmanned Aerial Vehicle (sUAV)

LASER SCANNING PROJECTS INVOLVED

1. Laramie Railroad Depot — Apr, 2024
2. Cheyenne High Plains Arboretum — Jul, 2024
3. Fort Laramie — Jun, 2025
4. Cheyenne Capitol Old — Aug, 2025

GRANTS AND PROPOSALS SUBMITTED

1. Wyoming NASA EPSCoR Faculty Research Grant on "Machine Learning-Driven 3D Spatial Analytics for Post-Wildfire Infrastructure and Landscape Risk Assessment using Laser Scanning." — Submitted: Mar, 2026
2. American Heritage Center (AHC) 2026 Teaching & Research Grant on "A Research Symposium on the Digital Preservation of the Apatosaurus at the University of Wyoming." — Submitted: Feb, 2026
3. Coalitions & Collaboratives (COCO) Action, Implementation & Mitigation Grant on "Predictive Vegetation Growth Modeling Near Power Lines Using Laser Scanning for Wildfire Fuel Reduction." — Submitted: Oct, 2025
4. National Forest Foundation 2025 Collaborative Capacity Program - Mini Grants on "Building Regional Capacity for Digital Preservation of At-Risk Heritage Infrastructure in Medicine Bow National Forest." — Submitted: Jul, 2025
5. National Science Foundation Advanced Sensing and Computation for Environmental Decision-making (ASCEND) Engine in Colorado and Wyoming Request for Information (Nascent Phase, Wildfire in Power/Water Systems) 2025 — Submitted: Jul, 2025
6. National Science Foundation CO-WY Engine on "Assessing and Monitoring the Carbon Sequestration Potential of Forested Areas in Colorado and Wyoming Using Advanced LiDAR and Drone Technologies" FY 2024-25 — Submitted: Aug, 2024

SOFTWARE

Cyclone Register 360 Plus, Autodesk Revit, Primavera P6, Microsoft Project, AutoCAD, ArcGIS, Civil3D